

Hospital Chatbot for Color Coded Clinical Pathways

Speaker Guide

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1 Overview

Title: Hospital Chatbot for Color Coded Clinical Pathways

Template: KNUST Beamer (Gabby.sty)

Duration: 20–30 minutes plus questions.

Core message: Patients present unstructured text X while vitals $\mathbf{v}$ are often unknown. The protocol still requires $\hat{c}$ and $C = f_{\text{SATS}}(\hat{c})$ before TEWS. The chatbot also binds C to pathway $P(C)$ and reduces acute-zone load (transshipment LP).

Key numbers:

- N=1000, skewed mix 5% / 15% / 30% / 25% / 25%
- Colour accuracy: 97.5% (975/1000)
- Level 1 recall: 94.0% (47/50)
- Green diversion: 500 patients (50%)
- Transshipment: cost 7150 $\rightarrow$ 2700 (62% reduction); acute load 900 $\rightarrow$ 500 (44%)

2 Section 1: Research gaps (1 slide)

2.1 Research gaps in prior work

Say: We aim to infer acuity from free-form chief complaints to reduce ED overcrowding and waiting times (Mitchell et al., 2023; Yilmaz, 2021). Each gap is one cited sentence from prior literature—not our project features.

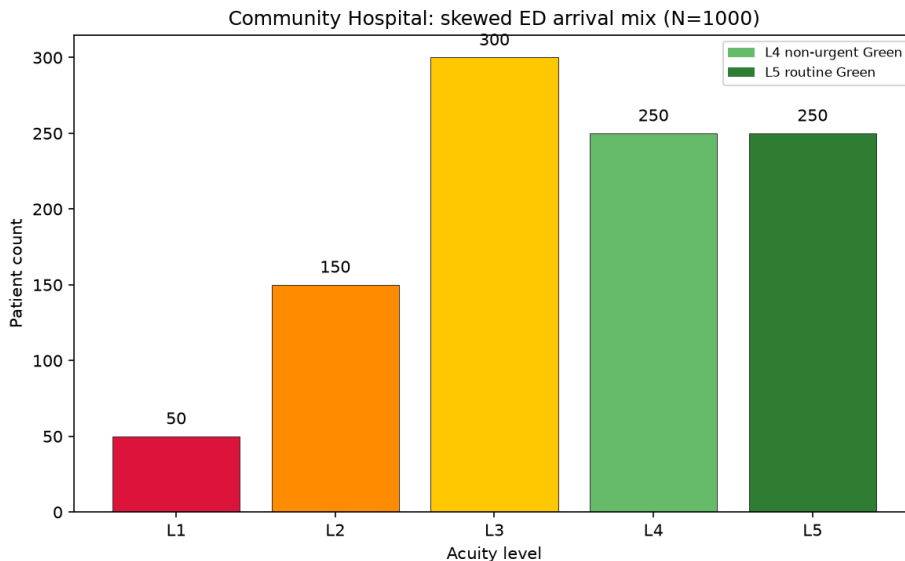
Walk all five items and name the references on slide:

1. Vitals-first bias — SATS2012, Rominski et al. 2014, Stewart et al. 2023.
2. Pathway discontinuity — Nurchis et al. 2025, Sundström et al. 2014.
3. Overcrowding without text pre-triage — Yilmaz 2021, Mitchell et al. 2023.
4. Generative vs constrained NLP — Stewart et al. 2023, Cheng et al. 2025.
5. Class imbalance — Stewart et al. 2023, Berkowitz et al. 2019.

Image-only gap slide removed. Own case-study numbers cite as **(This study, 2026)** on slides.

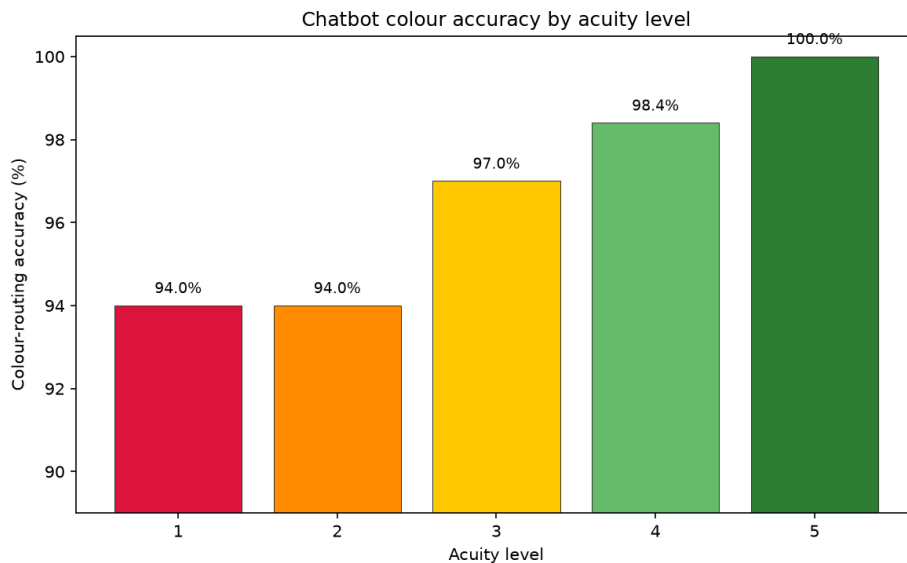
3 Section 2: Community Hospital (7 slides)

3.1 Acuity chart



Say: Skewed mix, not balanced. L4 and L5 bars are different shades; both map to SATS Green.

3.2 Accuracy chart



Say: 97.5% overall. Errors on adjacent levels and borderline cases (7% of rows flagged).

4 Section 3: Transshipment (4 slides)

Supplies: (50, 150, 300, 500). **Demands:** resus 50, majors 450, OPD 500.

Without chatbot: Green mis-routes to majors (acute load 900). With chatbot: flow 4→5→9 (acute load 500).

Network diagram: layer labels (Sources, Transshipment, Destinations) sit left of the node boxes. “Why transshipment fits triage” slide removed.

5 Section 4: NLP design (2 slides)

Fixed outputs: $\hat{c}$, C , $P(C)$. Contrast with generative LLMs (comparison table). BioBERT pipeline and future fusion slides removed; fusion remains in **triage-fusion/** documentation.

6 Section 5: Conclusion (2 slides)

Summary: four numbered takeaways (gaps, case study, transshipment, NLP design).

Limitations only: synthetic cohort, simulated eval, analytical transshipment model. No “next steps” slide.

References slide: final frame lists full bibliography (**references.bib**); compile with **bibtex main**.

7 Anticipated Q&A

Why synthetic data?

Skewed mix teaches realistic ED statistics; 8k holdout validates the model separately.

Why not ChatGPT?

Bounded output; no hallucinated urgency.

Why two greens on chart?

L4 non-urgent vs L5 routine; same SATS colour, different acuity levels.

Is TEWS included?

Not yet; see `triage-fusion/`.

8 Rebuild

```
python3 community-hospital/scripts/generate_synthetic_data.py
python3 community-hospital/scripts/run_chatbot_eval.py
community-hospital/.venv/bin/python community-hospital/scripts/transshipment_ed.py
community-hospital/.venv/bin/python community-hospital/scripts/analyze_case_study.py
cp community-hospital/analysis/figures/*.png community-hospital/slides/figures/
cd community-hospital/slides
pdflatex main.tex && bibtex main && pdflatex main.tex && pdflatex main.tex
cd ../guide && pdflatex presentation-guide.tex && pdflatex presentation-guide.tex
```